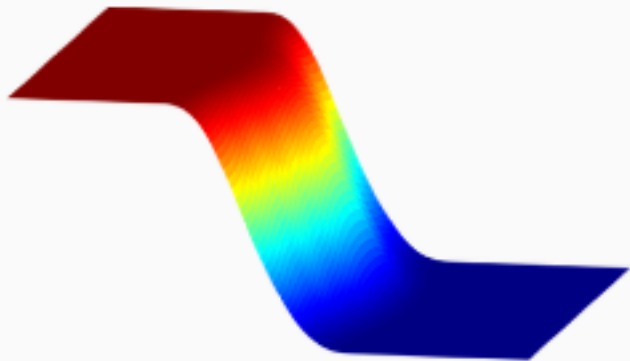
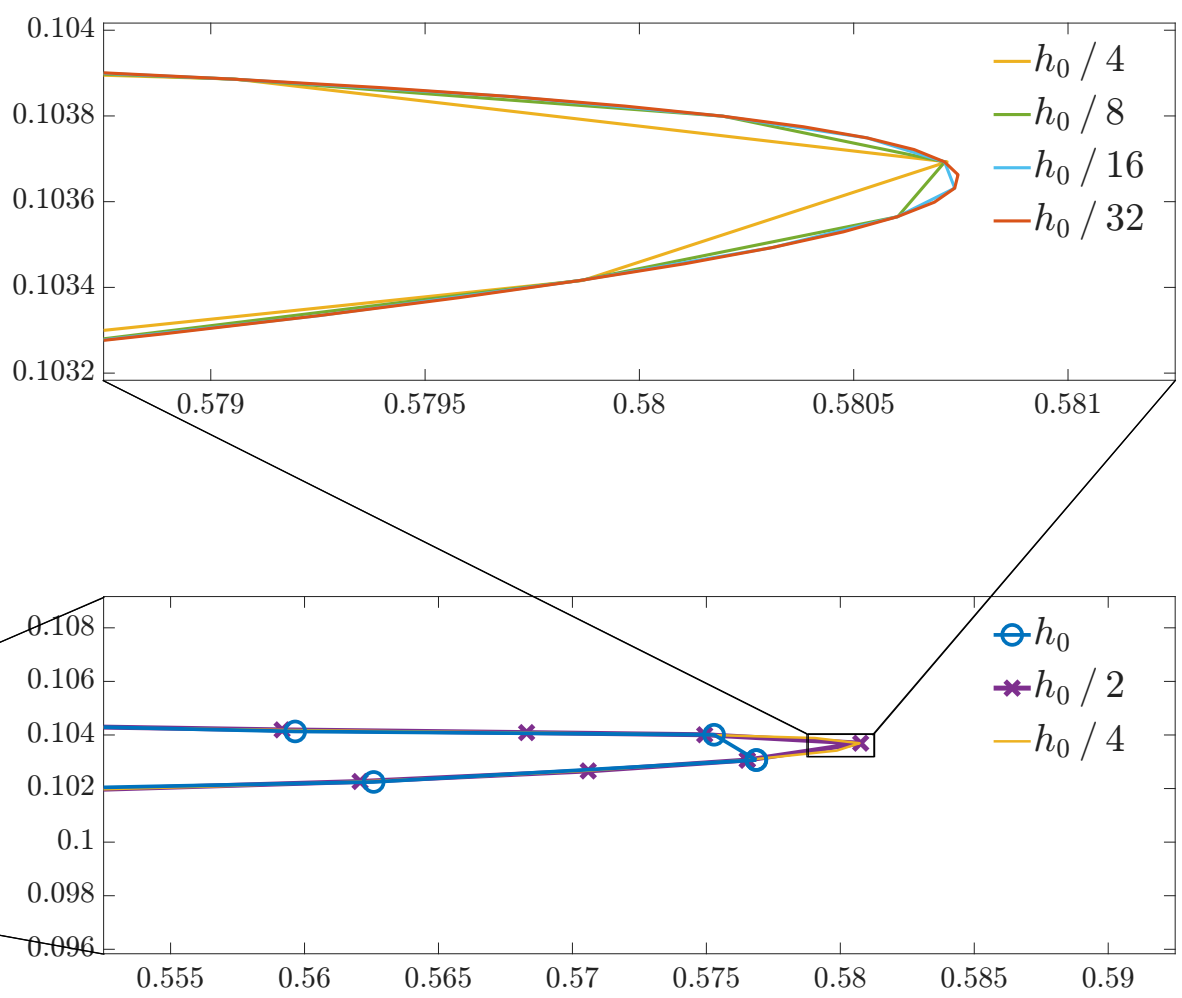
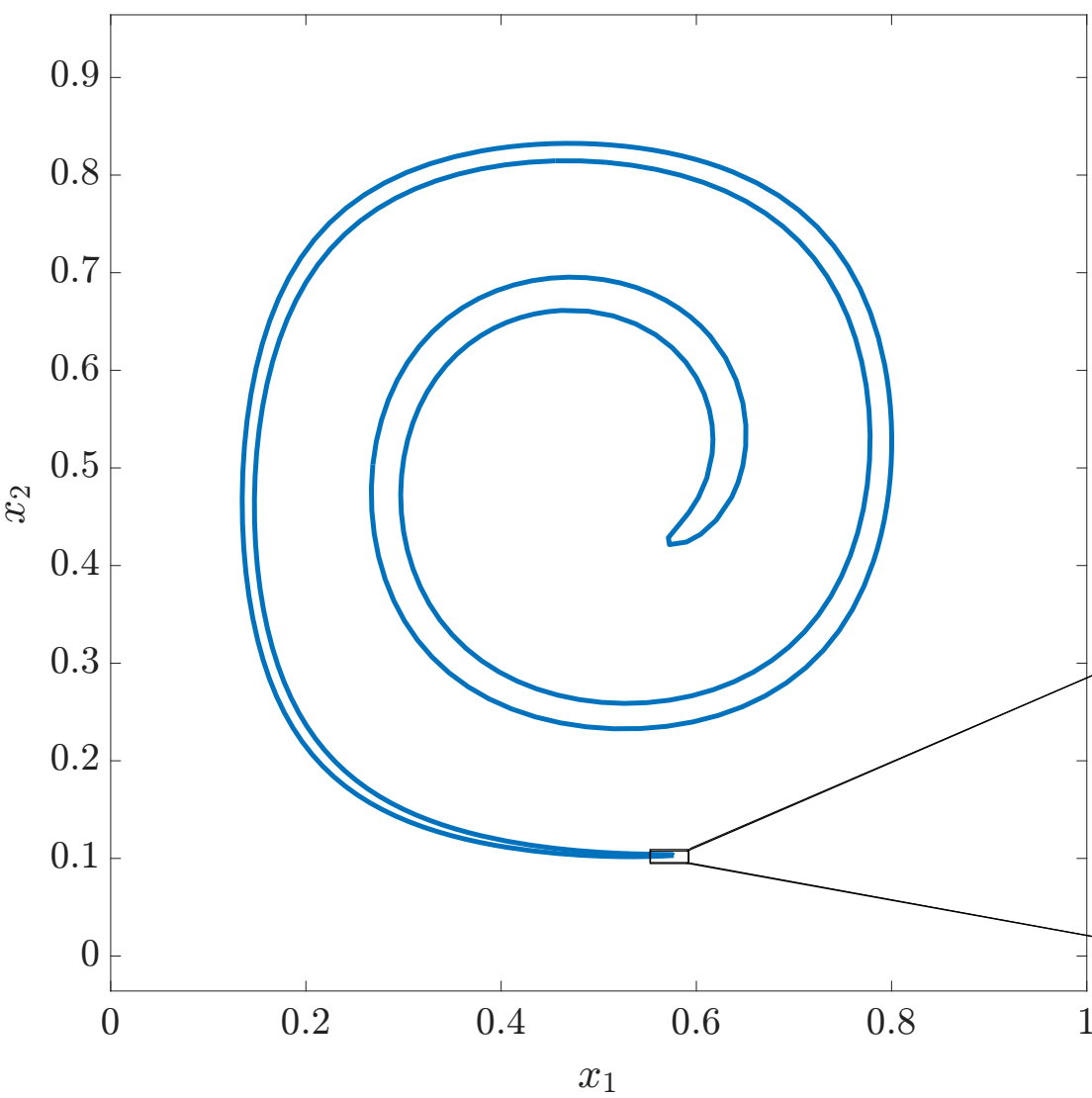
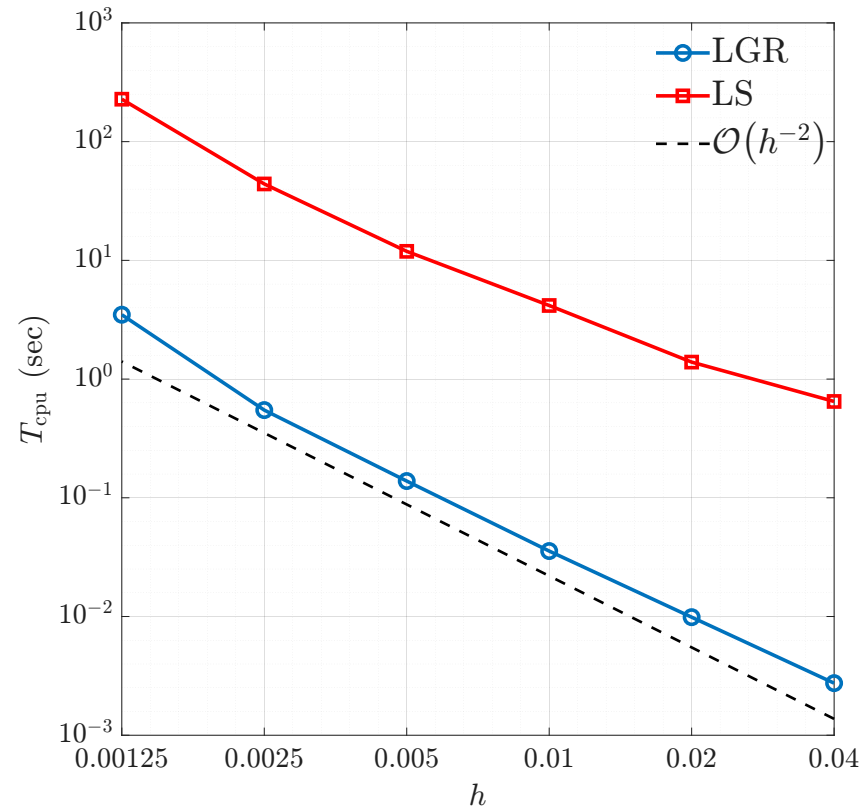
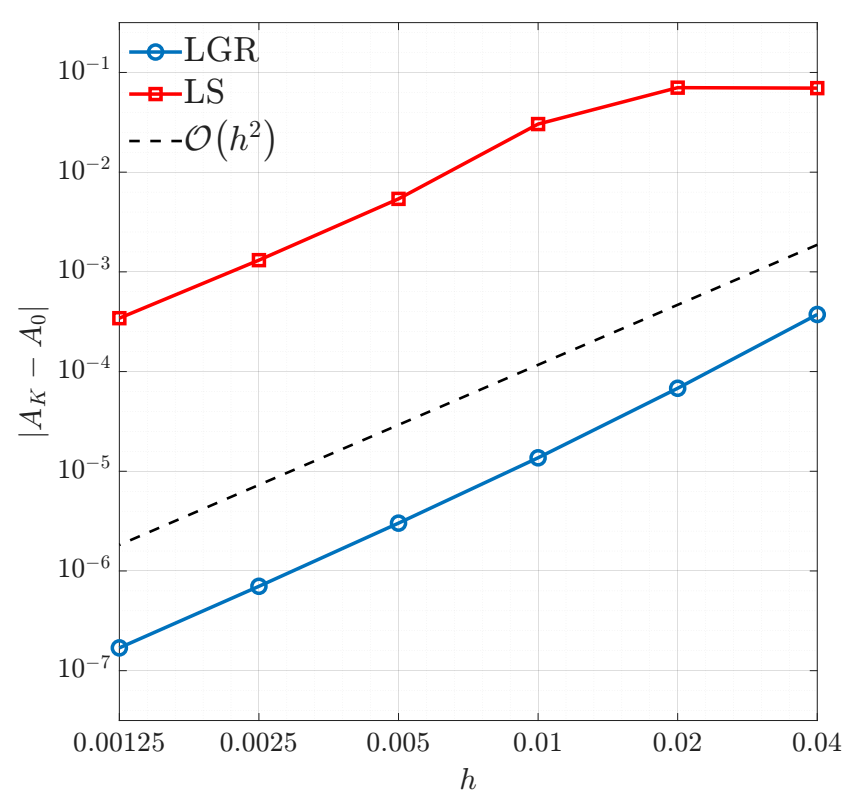




EFFICIENT AND ACCURATE INTERFACE TRACKING

EULERIAN CAPTURING

Interface is implicitly captured
by a level-set function on a
background mesh



LAGRANGIAN TRACKING

Interface is explicitly tracked
as a codimension—1
parametrized surface



$$\Gamma(t) = \left\{ \gamma(s, t) : s \in S^{d-1} \right\}$$

$$\Gamma(t) = \left\{ x \in \mathbb{R}^d : \phi(x, t) = 0 \right\}$$



$$\partial_t \phi + u \cdot \nabla \phi \equiv 0$$

$$\partial_t \gamma = u \circ \gamma$$



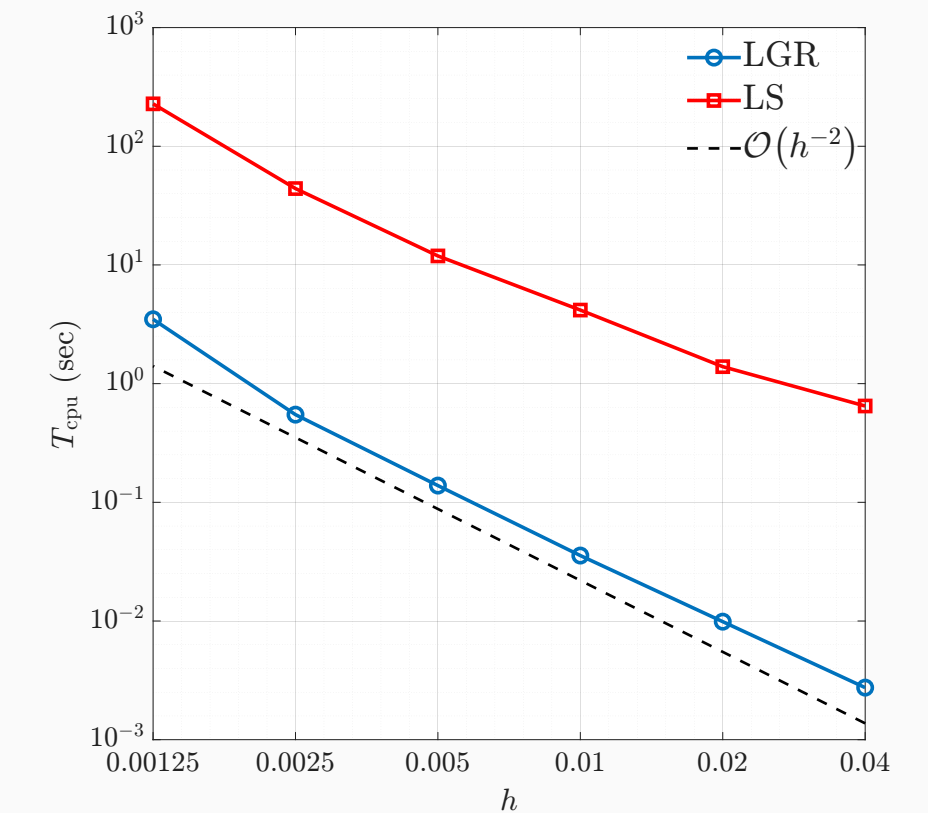
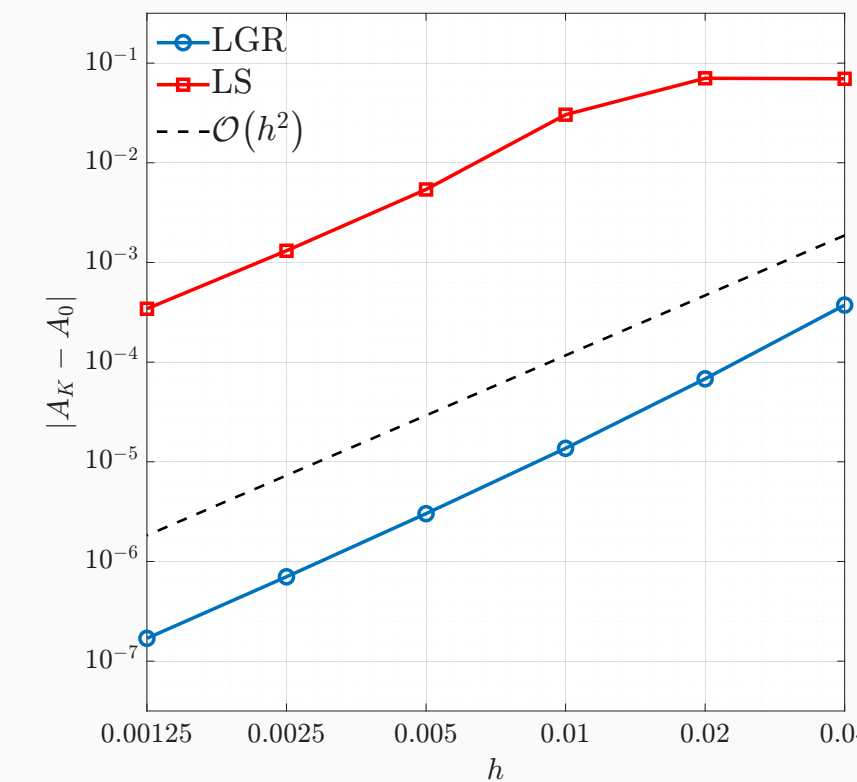
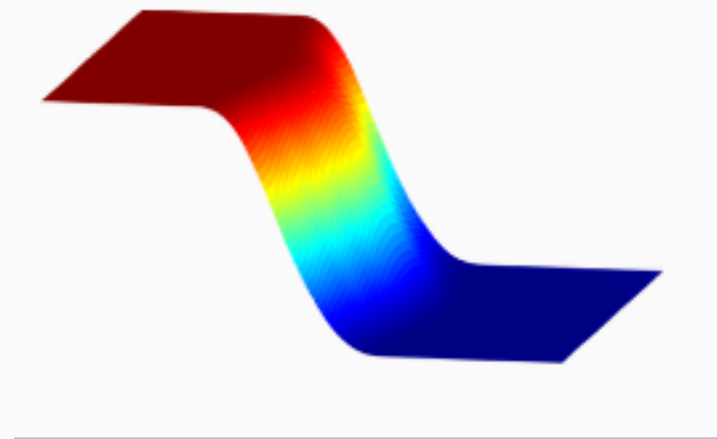
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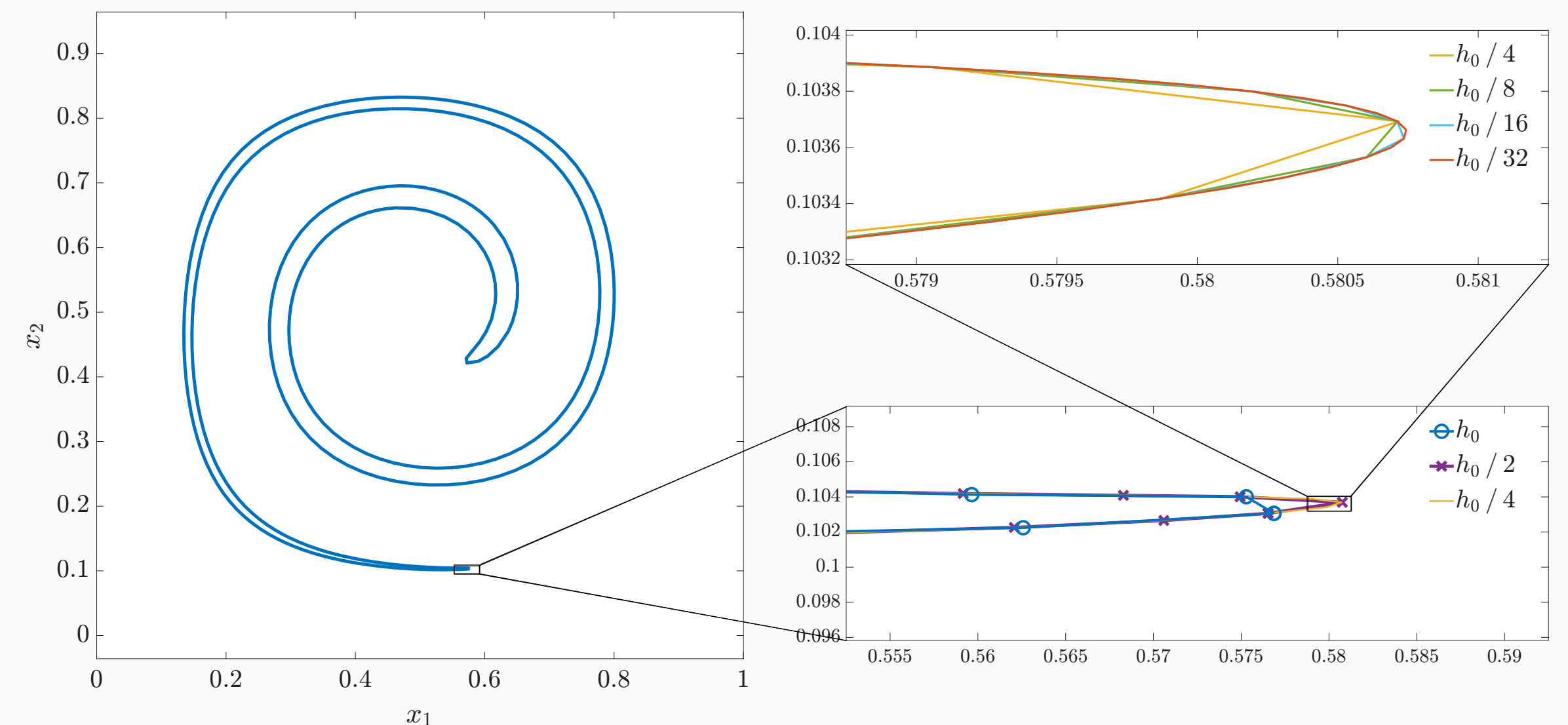


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MICROSCALE FILAMENTATION BY ALTERNATING SHEARS

